Teacher Wu Kuan | Digital SAT at a Glance — No Solution vs Infinitely Many

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1. The trap question

How do you tell no solution from infinitely many solutions?

2. The method

- Compare x and y coefficients: same ratio → same slope.
- Constants also in that ratio → same line → infinitely many.
- Constants not in that ratio → parallel → no solution.
- Coefficients not proportional → exactly one solution.

3. Parent check method

Have your child write the three ratios side by side (a_1/a_2, b_1/b_2, c_1/c_2) and compare.

Practice

1. $2x + 3y = 6$ and $4x + 6y = 12$: how many solutions?
2. $2x + 3y = 6$ and $4x + 6y = 10$: how many solutions?
3. $x + 2y = 5$ and $3x - y = 1$: solve.
4. $kx + 4y = 8$ and $3x + 2y = 5$ has no solution. Find k.
5. $3x - 2y = c$ and $6x - 4y = 10$ has infinitely many solutions. Find c.

Answers

1. Infinitely many
2. No solution
3. One solution: $x = 1$, $y = 2$
4. $k = 6$
5. $c = 5$

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